

22.1.4 Matching Considerations

The fact that performance depends on the matching of devices is another important aspect of differential amplifiers. Layout techniques can be used to minimize the first-order effects of mismatch due to oxide gradients and other process variations. Figure 22.18 shows how the diff-pair would be laid out in a *common-centroid* configuration (see Ch. 5). Common-centroid layout, as the name implies, constructs two devices symmetrically about a common center in the layout. This allows the two devices to cancel process gradients in both the x and y directions and exposes both devices to heat sources in an identical manner. Note how we've attempted to match the parasitic capacitances of the pair (on the gates and drains) and how we've minimized the capacitance at the sources (to keep the *CMRR* high at higher frequencies).

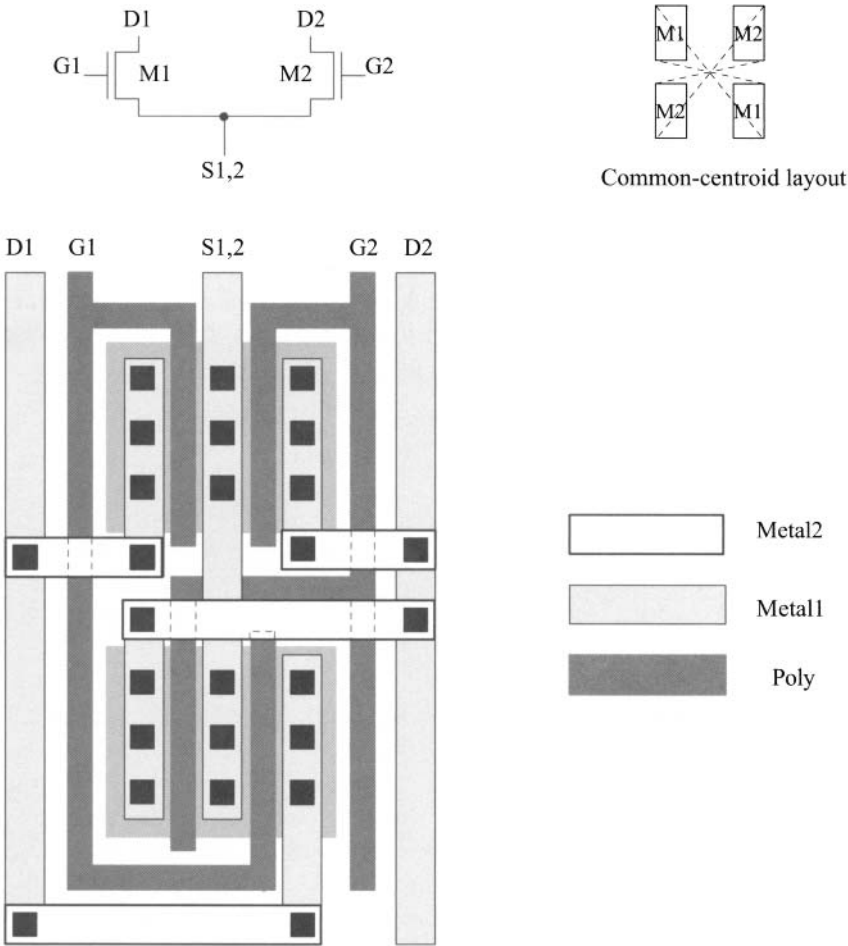


Figure 22.18 Common-centroid layout of a diff-pair.

The input-referred offset voltage of a diff-pair, resulting from mismatches in threshold voltage, geometries, and load resistances, can be determined with the help of Fig. 22.19. When the diff-pair is offset-free, connecting the diff-amp's inputs together causes the drain currents and drain voltages of M1 and M2 to be equal ($V_o = 0$). When the load resistors or M1 and M2 are mismatched, V_o is not zero. (We have mismatch and thus V_o is offset from its ideal value.). To drive V_o to zero, we apply a voltage difference between the gates of M1 and M2 (an input-referred offset).

$$V_{OS} = V_{GS1} - V_{GS2} = V_{THN1} + \sqrt{\frac{2I_{D1}}{\beta_1}} - V_{THN2} - \sqrt{\frac{2I_{D2}}{\beta_2}} \quad (22.28)$$

We can define the differences and averages of the threshold voltage, load resistance, and geometry as ΔV_{THN} ($V_{THN1} - V_{THN2}$), V_{THN} (average of V_{THN1} and V_{THN2}), ΔR_L ($R_{L1} - R_{L2}$), R_L (average of R_{L1} and R_{L2}), $\Delta(W/L)$ [$(W_1/L_1) - (W_2/L_2)$] and (W/L) [average of (W_1/L_1) and (W_2/L_2)]. Thus, we can make the appropriate substitutions into Eq. (22.28), requiring $I_{D1}R_{L1} = I_{D2}R_{L2}$, with the result:

$$V_{OS} = \Delta V_{THN} + \frac{V_{GS} - V_{THN}}{2} \cdot \left[\frac{-\Delta R_L}{R_L} - \frac{\Delta(W/L)}{(W/L)} \right] \quad (22.29)$$

The threshold voltage mismatch must be reduced using layout techniques. Mismatches resulting from unequal geometry and differences in the load resistance can be reduced by designing with a small V_{GS} (V_{GS} close to V_{THN}). This should be compared with Eq. (20.8), which shows that using a small V_{GS} in a current mirror results in a large mirrored current difference.

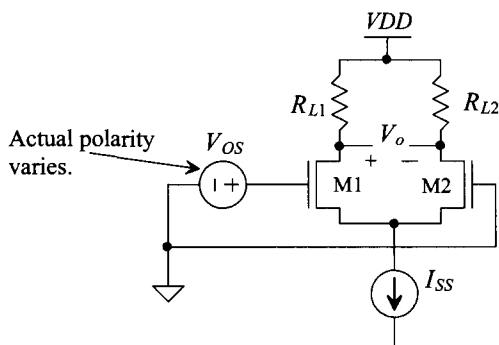


Figure 22.19 Determining diff-amp input offset voltage.

Input-Referred Offset with a Current Mirror Load

Suppose the diff-amp with current mirror load, Fig. 22.12, has an offset (the devices aren't perfectly matched). What this means is that the drain potential of M4 (M2) is not equal to the drain potential of M3 (M1 and most likely the drain currents in M1 and M2 aren't equal as well). The input-referred offset would then be the voltage difference on the gates of M1 and M2 needed to drive the output voltage to the correct value. Using Eq. (22.22), we can write the input-referred offset in terms of the difference-mode gain as

$$|V_{OS}| = \frac{V_{o,ideal}}{|A_d|} = \frac{V_{o,ideal}}{g_m \cdot (r_{o2} || r_{o4})} \quad (22.30)$$

which indicates that to reduce the input-referred offset voltage we simply need to design the diff-amp with large gain. This is the same result as that of Eq. (22.29), if we pause and think about it. To reduce the offset, we said we design with a small V_{GS} . If we hold the diff-amp's bias current constant and reduce V_{GS} (by increasing the widths of M1 and M2), we are increasing g_m .

Finally, although not clearly indicated by Eq. (22.30) as it is in Eq. (22.29), a mismatch in the diff-pair's threshold voltage can only be reduced by layout techniques (e.g., using common-centroid layout). Using large devices (larger-width MOSFETs), reduces variation in device characteristics and leads to better matching.

22.1.5 Noise Performance

The diff-amp with noise sources is seen in Fig. 22.20. The noise from M6 feeds equally into M1 and M2 (see Fig. 21.44 for comments about noise contributions from M5) and thus, because of the current mirror action in M3 and M4, ideally doesn't affect the output voltage. Therefore, we will ignore noise from M6 in the following discussion.

The diff-amp's output noise power spectral density (PSD) is given by

$$V_{noise}^2(f) = (I_{M1}^2(f) + I_{M2}^2(f) + I_{M3}^2(f) + I_{M4}^2(f)) \cdot (r_{o2} || r_{o4})^2 \quad (22.31)$$

The input-referred noise PSD is then

$$V_{noise}^2(f) = \frac{V_{noise}^2(f)}{A_d^2} \quad (22.32)$$

Substituting in the difference-mode gain of the diff-amp, Eq. (22.22), we get

$$V_{noise}^2(f) = \frac{I_{M1}^2(f) + I_{M2}^2(f) + I_{M3}^2(f) + I_{M4}^2(f)}{g_m^2} \quad (22.33)$$

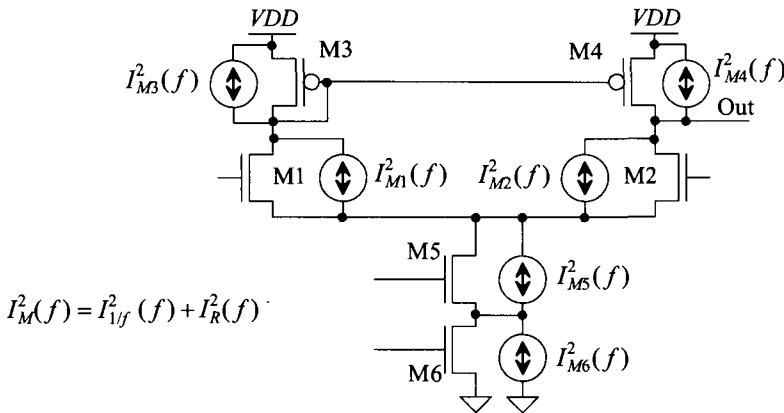


Figure 22.20 Noise model for the diff-amp.

By maximizing the transconductance, g_m , of the diff-pair, we can reduce the input-referred noise. This statement alone is a little misleading. As we saw in Eqs. (9.63)–(9.65), increasing the drain current increases the noise PSD. So to reduce the input-referred noise, we would want to maximize the transconductance by increasing the diff-pair's width (make M1 and M2 wide), not by increasing I_{SS} .

22.1.6 Slew-Rate Limitations

Like the class A amplifiers studied in the last chapter, the basic diff-pair also exhibits slew-rate limitations. As seen in Fig. 22.21, when either M1 or M2 turns off, the total current available to charge a load capacitance is I_{SS} . The slew-rate is then (see also Eq. (21.38))

$$\text{slew rate} = \frac{dV_{out}}{dt} = \frac{I_{SS}}{C_L} \quad (22.34)$$

For high-speed, we need a large bias current. However, this causes excess power dissipation. Once again, to get high-speed (large current drive capability) with low quiescent power dissipation, let's look at a class AB configuration.

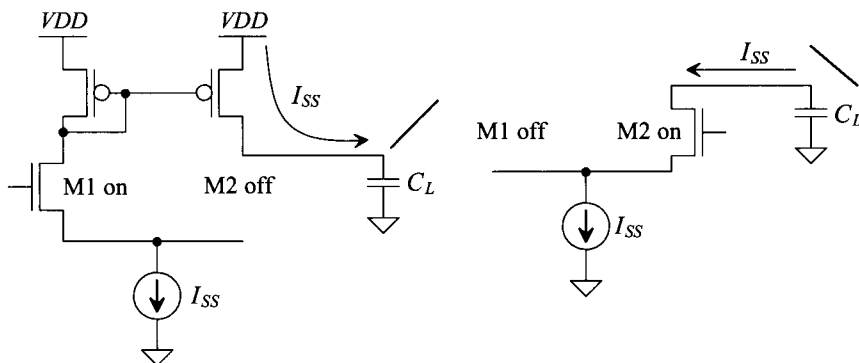


Figure 22.21 Slew-rate limitations in diff-amps.

22.2 The Source Cross-Coupled Pair

The source cross-coupled pair is shown in Fig. 22.22. This configuration is very useful in practical circuit design because it can eliminate slew-rate limitations. A diff-amp built with the source cross-coupled pair can operate in the class AB mode with significant output drive current. Assuming that both inputs to the pair are connected to a (the same) voltage within the pair's common-mode input range and that all NMOS are sized the same and that all PMOS are sized the same, a current I_{SS} flows in all the MOSFETs in the circuit. Note that M11, M21, M31, and M41 are simply behaving as biasing batteries. Their gate-source voltages are mirrored by M1–M4 to set the DC operating current. Using the parameters in Table 9.1 with the gates of M1 and M2 (v_{n1} and v_{n2}) at 3.5 V and I_{SS} of 20 μA , the gate of M3 is at a constant potential that we will label $v_{n2} - V_{bias}$ (or $V_{bias} = V_{GS21} + V_{SG31} = 2.2$ V neglecting body effect).

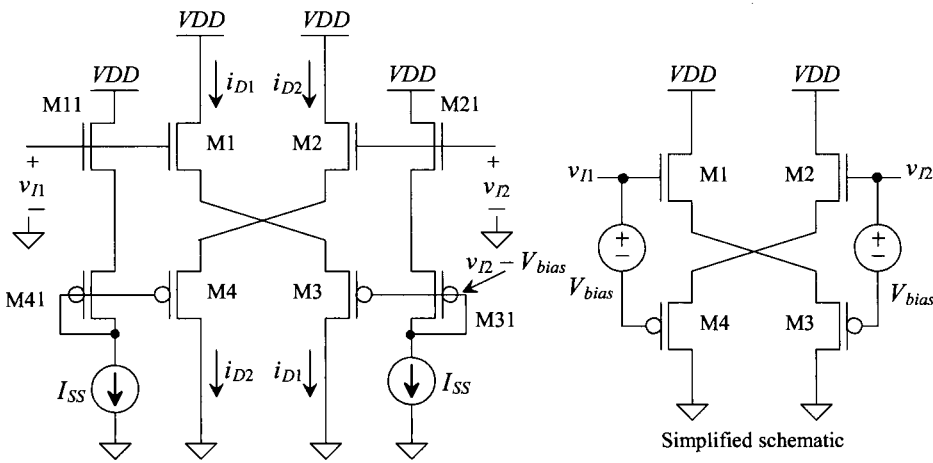


Figure 22.22 Source cross-coupled differential amplifier with NMOS input devices.

Operation of the Diff-Amp

An important characteristic of the source cross-coupled differential amplifier is that as v_{DI} is increased, i_{D1} continues to increase and i_{D2} shuts off, while the opposite is true when v_{DI} decreases. In other words, the amplifier is operating class AB where neither of the output currents is zero as long as their magnitudes remain less than I_{SS} . Looking at the simplified schematic in Fig. 22.22, if v_{I1} increases, then M1 turns on. Because the gate potential of M3 is constant, it turns on as well. In other words, an increase in v_{I1} causes the gate-source voltages of both M1 and M3 to increase (i_{D1} increases). At the same time, the gate potential of M4 increases. This cause both M4 and M2 to shut off (i_{D2} shuts off).

Example 22.8

Simulate the operation of the diff-amp in Fig. 22.22 using the parameters from Table 9.1.

The simulation results are seen in Fig. 22.23. Note that when both inputs are at 3.5 V, the output currents are equal. This figure should be compared to Fig. 22.3.

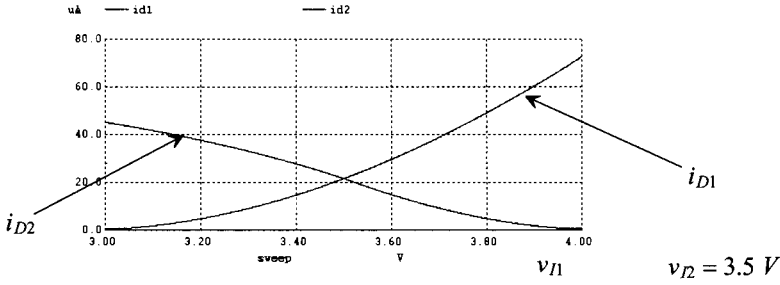


Figure 22.23 The output currents of the diff-amp in Fig. 22.22.

Note how in Fig. 22.3, the output currents flatten out above a maximum input voltage. Here the current continues to increase. ■

After looking at the results of this example, we might wonder: why use such a complicated diff-amp if the output current is only three times more than what we get with the basic diff-amp discussed in the first section? We can always just scale up the current in a basic diff-amp to get more drive. The benefits of the source cross-coupled diff-amp are seen in special-purpose, *low-power* design. In low-power design, the gate source voltages are close to the threshold voltages (and so low power equates to low speed, see Eq. [9.36]). Consider the amplifier and simulation results seen in Fig. 22.24. Here we've increased the widths of the amplifying devices and reduced their bias current (by sizing up the lengths of the biasing current sources as seen in the figure). When the two inputs are the same voltage, the quiescent current that flows in the diff-amp is a few microamps. However, if one input gets a little larger than the other input, the drain-current increases dramatically. In a low-power design, this topology would be useful to quickly charge a capacitor while burning little quiescent power.

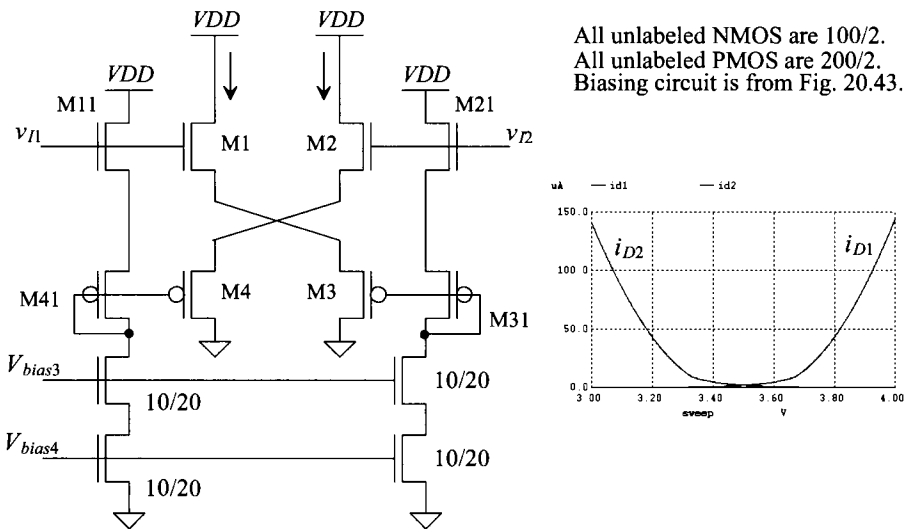


Figure 22.24 Lowering the power in the source cross-coupled diff-amp.

Input Signal Range

The drawback of the source cross-coupled diff-amp is the limited input range. The PMOS version of the source cross-coupled differential amplifier is shown in Fig. 22.25. Normally the version (PMOS or NMOS) of this differential amplifier depends on the input signal range. The NMOS version has the best positive input signal range, while the PMOS version has the best negative input signal range. For centering the input signal range the topology seen in Fig. 22.26 can be used. Again, both an NMOS V_{GS} and a PMOS V_{SG} are used to bias the cross-coupled diff-amp. Note that the biasing devices

Figure 22.27 Source cross-coupled differential amplifier, with current source loads.

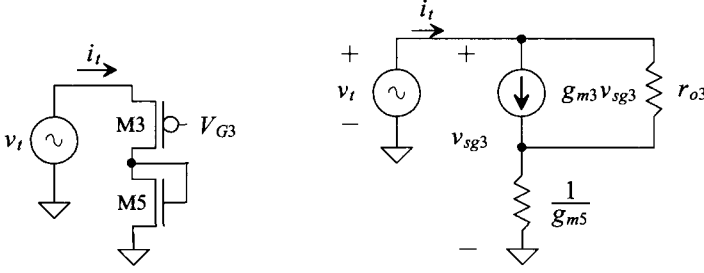


Figure 22.28 (a) Simplified schematic and (b) small-signal model.

The voltage gain from v_{s1} to v_{g9} is the same form as a common gate amplifier. This gain is

$$\frac{v_{g9}}{v_{s1}} = g_{m3} \frac{1}{g_{m5}} \quad (22.38)$$

And finally, the gain from the gate of M9 to the output is simply $-g_{m9}(r_{o9} || r_{o10})$, and

$$\frac{v_o}{v_{i1}} = -\frac{g_{m1} \frac{1}{g_{m5}}}{1 + g_{m1} \frac{1}{g_{m3}}} g_{m9}(r_{o9} || r_{o10}) \text{ for } v_{i2} = 0 \quad (22.39)$$

Using the same type of analysis, we find that the expression from v_{i2} to the output is

$$\frac{v_o}{v_{i2}} = \frac{g_{m2} \frac{1}{g_{m6}}}{1 + g_{m2} \frac{1}{g_{m4}}} g_{m10}(r_{o9} || r_{o10}) \text{ for } v_{i1} = 0 \quad (22.40)$$

Superposition can now be used to determine the total effect of each input on the output. Therefore, if M1 and M2 are matched, as are M3 and M4, and if M5 and M6 are designed so that their g_m s are equal as are M7 and M8 (respectively), then the differential gain can be expressed as

$$v_o = (v_{i2} - v_{i1}) \frac{2 \cdot g_{m1,2} \frac{1}{g_{m5,6}}}{1 + g_{m1,2} \frac{1}{g_{m3,4}}} g_{m9,10}(r_{o9} || r_{o10}) \Rightarrow \frac{v_o}{v_{di}} = \frac{2 \cdot g_{m1,2} \frac{1}{g_{m5,6}}}{1 + g_{m1,2} \frac{1}{g_{m3,4}}} g_{m9,10}(r_{o9} || r_{o10}) \quad (22.41)$$

Input Signal Range

The positive-input, common-mode range (CMR) is determined in exactly the same manner as in the source coupled diff-amp of the first section. The positive CMR is given by

$$v_{CMMAX} \approx VDD - \sqrt{\frac{2I_{SS}}{\beta_6}} = VDD - V_{DS,sat} \quad (22.42)$$

while the negative CMR is given by

$$v_{CMMIN} = \sqrt{\frac{2I_{SS}}{\beta_1}} + V_{THN} + \sqrt{\frac{2I_{SS}}{\beta_3}} + \sqrt{\frac{2I_{SS}}{\beta_5}} + V_{THN} \quad (22.43)$$

An alternative current source (current mirror) load configuration is shown in Fig 22.29. The gain of this configuration is given by

$$A_d = \frac{v_{out}}{v_{i1} - v_{i2}} = 2 \cdot \frac{r_{o2} || r_{o6}}{\frac{1}{g_{m1}} + \frac{1}{g_{m3}}} \quad (22.44)$$

assuming $g_{m1} = g_{m2}$ and $g_{m3} = g_{m4}$.

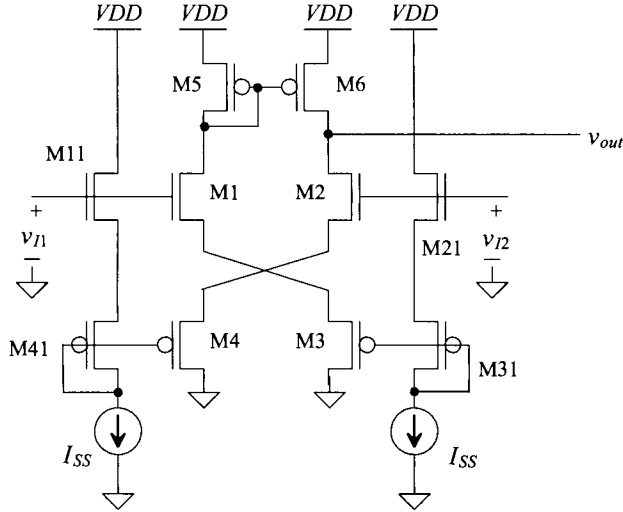


Figure 22.29 Alternative current source load configuration.

22.3 Cascode Loads (The Telescopic Diff-Amp)

The differential amplifier with current source load of Fig. 22.6 gave a voltage gain of $g_m(r_{o2} || r_{o4})$. For many applications, this gain may be too low. Using a cascode current source load in place of M3/M4, results in a gain of approximately $g_m r_{o2}$ (the output resistance of the load becomes much larger than the resistance looking into the drain of M2). This modest improvement in gain can be increased by also cascoding M1 and M2. Figure 22.30 shows the resulting circuit configuration (sometimes called a *telescopic diff-amp*). The MOSFET M6 is selected so that its V_{GS} keeps M1/M2 and MC1/MC2 in the saturation region. The gain of this configuration is given by

$$A_d = g_{m1} \cdot (R_{into\ DC2} || R_{into\ DC4}) \quad (22.45)$$

The resistance looking into the drain of MC2, assuming M2 and MC2 are the same size, is given by

$$R_{into\ DC2} \approx g_{m2} \cdot r_{o2}^2 \quad (22.46)$$

and the resistance looking into the drain of MC4, again assuming MC4 and M4 are the same size, is given by

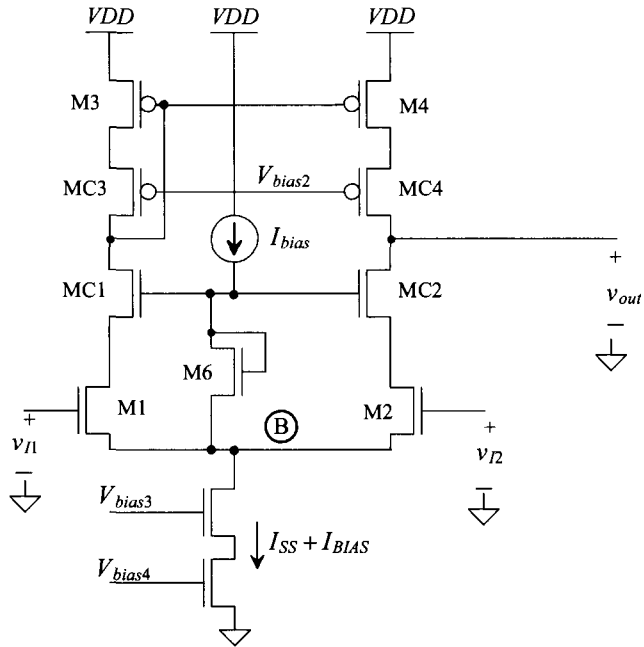


Figure 22.30 Cascode differential amplifier.

$$R_{into\ DC4} \approx g_{m4} \cdot r_{o4}^2 \quad (22.47)$$

The gain of the cascode differential amplifier may be written as

$$A_d = g_{m1} (g_{m2} r_{o2}^2 \parallel g_{m4} r_{o4}^2) \quad (22.48)$$

The main drawback of using the cascode differential amplifier is the reduction in positive common-mode range. In the negative direction, V_{CMMIN} is identical to the source-coupled diff-amp discussed in Sec. 22.1. The positive common-mode signal is limited by the additional circuitry required by the cascode loads. Notice that V_{CMMAX} in this case is limited by the amount of voltage necessary to keep M1, MC1, MC3, and M3 in their respective saturation regions.

An interesting issue concerning this circuit is the DC feedback through M6. Because the current through M6 is constant, V_{GS6} will be constant. Normally, an increase in the common-mode voltage on the gates of M1 and M2 causes their drain voltages to decrease. However, because the currents through M1 and M2 are constant, the common source node, marked as node B in Fig. 22.30, increases with the increasing input voltage. This causes the gate potential of MC1 and MC2 to increase. The result is that the drain voltages of M1 and M2 go up (since the gate-source voltages of MC1 and MC2 are constant). This feedback action attempts to keep M1 and M2 from going into nonsaturation. The maximum common-mode voltage, V_{CMMAX} , will be limited by the output voltage and by MC1/MC2 going into nonsaturation.